

Coding & Solution

Date	30 June, 2026
Team ID	
Project Name	Hydra Shield: An Intelligent Flood Prediction and Early Warning System (Rising Waters)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the implementation quality of the HydraShield: An Intelligent Flood Prediction and Early Warning System project. The project is developed using Machine Learning and Flask to predict flood risks based on environmental and weather parameters. The code follows Python coding standards, is modular, well-documented, and satisfies the functional requirements of the proposed system.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/jagannagari/AI-ML-SmartbridgeProject-Credit-Card-Approval-Prediction-
Programming Language(s)	Python 3.11, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Joblib, Matplotlib
Key Features Implemented	• Data preprocessing and cleaning using Pandas • Feature selection and preprocessing for flood prediction • Model training using Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost • Best-performing model saved using Joblib • Flask-based web application for real-time flood prediction • Interactive user interface for entering environmental parameters • Instant flood risk prediction with AI-based recommendations
Pending / Incomplete Features	• Live weather API integration • SMS/Email alert notification system • Interactive GIS/Map visualization • IBM Cloud deployment (Future Enhancement)

Setup / Run Instructions	1. Install Python libraries using pip install -r requirements.txt. 2. place model.pkl in the project folder. 3. Run python app.py. 4. Open http://127.0.0.1:5000 in a web browser. 5. Enter applicant details and click Predict to view the result.
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Code Quality Checklist

S.No	Criteria	Status(yes/no)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The HydraShield: An Intelligent Flood Prediction and Early Warning System follows modern software engineering practices by maintaining a modular architecture, reusable code components, and comprehensive documentation. Multiple machine learning algorithms are trained and evaluated to identify the best-performing model for flood prediction. The selected model is integrated into a Flask-based web application that provides real-time flood risk assessment and AI-generated recommendations. The project is designed to be scalable and can be enhanced further by integrating live weather APIs, GIS-based flood visualization, cloud deployment, notification services, and historical prediction analytics.